

Arnav Devalapally

Perception ML Research Engineer

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EDUCATION

M.S. Computer Science - University of Michigan, Ann Arbor, USA GPA 4.0/4.0 | Aug 2025 - Dec 2026
• **Coursework:** Advanced Computer Vision, Self-driving cars (Fall 26), Robot Kin. & Dyn., Control Systems

B.Tech. AI & Data Science (Hons.) - KL University Hyderabad, India GPA 3.9/4.0 | Aug 2021 - May 2025

PUBLICATIONS

Source Models Leak What They Shouldn't: Unlearning Zero-Shot Transfer in Domain Adaptation Through Adversarial Optimization - CVPR 2026 (first author) [arxiv](#), [github](#)
• Formalized a setting and developed a method that adapts a model to a new environment while unlearning sensitive classes: <1% sensitive-class leakage with 95%+ accuracy across 7 benchmarks (including 2 real-world).

A Simple Machine Unlearning Approach via Elastic Weight Consolidation - Springer CCIS ICRTAET 2023 (first author)
• Reframed unlearning as continual learning via EWC: held ~80% retained accuracy vs. ~10% collapse.

EXPERIENCE

Perception ML Research Engineering Intern - Parsons Corporation May 2026 - Present
• Build perception systems that detect 15+ roadway assets and defects (e.g., potholes, cracks, worn and fallen signs) from raw camera and LiDAR data across two client deployments, with broader product plan.
• Own an end-to-end road-sign pipeline: detection, multi-view association and clustering, prediction pooling, and zero-shot classification leveraging foundational models: Grounding DINO, DINOv2, CLIP, and perception methods.

Research Assistant, Perception & Simulation - University of Michigan | Advisor: Alan Papalia Mar 2026 - May 2026
• Built a detailed Isaac Sim environment with a payload-equipped Boston Dynamics Spot (Ouster OS0 LiDAR, ZED 2i stereo, VN-100 IMU) to generate synthetic training data.

Computer Vision Researcher - IIT Hyderabad | Advisor: Vineeth N. Balasubramanian May 2024 - Feb 2026 (Part-time)
• Designed the domain-adaptation framework above; benchmarked leakage with membership-inference attacks: 22% lower attack success than the source model, beating 5+ baselines on DomainNet-126 benchmark.
• Ran 500+ experiments on a SLURM cluster to produce 30+ tables of ablations and baselines.

ML Systems Engineer, FinTech - PXP Financial & Payintelli Feb 2025 - Aug 2025
• Built fraud-detection models on 1M+ transactions, raising fraud recall 50% while keeping false positives under 1% with calibrated thresholds.
• Served models at under 100 ms latency across 500k+ transactions.

PROJECTS

Maverick, Autonomous Robot - UofM IGCV Competition Team Jan 2026 - Apr 2026
• Fused two ZED 2i stereo cameras into one occupancy grid, removing blind spots and expanding coverage ~20%.
• Cut perception latency from ~160 ms to ~30 ms with GPU acceleration (PyTorch + CUDA).
• Built a calibration UI for tuning the 3-DoF camera poses used in the grid fusion.

ACHIEVEMENTS

- 15+ robotics competition titles (FLL, WRO, etc), including 4x Robofest World Championships (100+ teams).
- President's Silver Medalist & full merit scholarship, KL University (1 of 2 awardees).

SKILLS

ML & Perception: Domain Adaptation / Domain Generalization, 3D Object Detection, Semantic SLAM, Sensor Fusion, BEV & Occupancy Mapping, Vision-Language Models (CLIP, DINOv2, Grounding DINO), NeRF, **3D Vision:** ICP, RANSAC, Pose Calibration, Point Clouds, Stereo, **Frameworks:** PyTorch, CUDA, OpenCV, Open3D, PCL, tllib, **Robotics & Sim:** ROS2 Humble, NVIDIA Isaac Sim, Gazebo, Rviz2, **Deployment & MLOps:** AWS (SageMaker), AzureML, Docker, SLURM, CI/CD (GitHub Actions), pytest, Linux, Git, API design, PySpark, TensorRT, **Languages:** Python, C++, Bash, MATLAB **Hardware:** Jetson Orin, Ouster OS0 LiDAR, ZED 2i, VectorNav VN-100 IMU